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The Geometry of Thought: How Scale Restructures Reasoning in Large Language Models

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Abstract

Scale does not uniformly improve reasoning—it restructures it. Analyzing 25,000+ chain-of-thought trajectories across four domains (Law, Science, Code, Math) and two scales (8B, 70B parameters), we discover that neural scaling laws trigger domain-specific phase transitions rather than uniform capability gains.

Legal reasoning undergoes *Crystallization*: 45% collapse in representational dimensionality (d_{95}: 501 $\rightarrow$ 274), 31% increase in trajectory alignment, and 10$\times$ manifold untangling. Scientific and mathematical reasoning remain *Liquid*—geometrically invariant despite 9$\times$ parameter increase. Code reasoning forms a discrete *Lattice* of strategic modes (silhouette: 0.13 $\rightarrow$ 0.42).

This geometry predicts learnability. We introduce Neural Reasoning Operators—learned mappings from initial to terminal hidden states. In crystalline legal reasoning, our operator achieves 63.6% accuracy on held-out tasks via probe decoding, predicting reasoning endpoints without traversing intermediate states.

We further identify a universal oscillatory signature (coherence ≈ -0.4) invariant across domains and scales, suggesting attention and feedforward layers drive reasoning through opposing dynamics.

These findings establish that the cost of thought is determined not by task difficulty but by manifold geometry—offering a blueprint for inference acceleration where topology permits.

1 Introduction

The scaling hypothesis has become the central organizing principle of modern deep learning: larger models, trained on more data, yield predictably better performance across an expanding frontier of tasks. Neural scaling laws formalize this intuition, demonstrating that loss decreases as a power law in model parameters, dataset size, and compute budget [25, 27]. For reasoning tasks specifically, chain-of-thought (CoT) prompting has revealed that scale unlocks qualitatively new capabilities—arithmetic, symbolic manipulation, and multi-hop inference—that smaller models cannot perform regardless of prompting strategy [57].

Yet the scaling hypothesis, as typically articulated, is silent on a crucial question: *how does scale improve reasoning?* The default assumption is that larger models simply "know more"—that additional parameters encode additional facts, patterns, or heuristics that accumulate to produce better outputs. Under this view, scaling is purely quantitative: more of the same, executed more reliably.

This paper presents evidence for a sharply different picture. We demonstrate that scale does not merely improve reasoning—it restructures it, triggering qualitative geometric reorganizations in how models represent the reasoning process itself. Moreover, these reorganizations are domain-dependent: the same 9$\times$ increase in parameters induces dramatically different effects depending on what the model is reasoning about. This paper asks a deliberately sharp question:

Is reasoning inherently a path, or can it be treated as an endpoint computation?

The question is not about whether intermediate text is useful, nor whether verbalized CoT is a faithful transcript of internal computation. Rather, it asks whether the internal evolution of model representations during reasoning exhibits sufficient geometric regularity that the *endpoint* of that evolution can be predicted without explicitly simulating every intermediate step.

1.1 The dynamical view: reasoning as trajectories

We adopt a dynamical systems perspective on transformer computation. Let $h_t \in \mathbb{R}^d$ denote the final-layer hidden state at token position t during chain-of-thought generation. A *reasoning trajectory* is the sequence

$$\tau = (h_0, h_1, \ldots, h_T) \tag{1}$$

traced through representation space as the model generates intermediate reasoning text. While the underlying computation is history-dependent (attention conditions on all previous tokens), the observed hidden states are points in a fixed-dimensional space whose geometric properties can be measured and compared.

This perspective transforms abstract questions about "reasoning capability" into concrete questions about manifold structure: What is the effective dimensionality of reasoning trajectories? Do trajectories align toward canonical pathways or diffuse through high-dimensional space? Do distinct reasoning strategies cluster into separable modes? How do these geometric properties change with scale?

Prior work in mechanistic interpretability has begun characterizing such structure. Li et al. (2025) formalize a "reasoning manifold" and use geometric deviation to diagnose failures [32]. Carson and Reisizadeh (2025) model sentence-level trajectories as drift-diffusion processes with regime switching [10]. Marin (2024) proposes phase-space analysis of multi-hop reasoning [38]. However, these approaches are primarily diagnostic—they identify geometric signatures *post hoc* but do not systematically investigate how geometry varies across domains or scales.

We extend this program by treating trajectory geometry as an independent variable. Concretely, we extract hidden-state trajectories during chain-of-thought generation across four reasoning domains (Law, Science, Code, Math) and two model scales (Llama-3-8B-Instruct and Llama-3.1-70B-Instruct), quantify their geometric properties using a standardized measurement suite, and test whether the resulting structure supports learned operators that can predict terminal hidden states without traversing full trajectories.

1.2 A geometric measurement suite

We introduce four complementary metrics to characterize reasoning geometry:

- **Global dimension (d_{95}):** the number of principal components required to capture 95% of variance in the trajectory point cloud—the effective linear dimensionality of the reasoning manifold.
- **Intrinsic dimension (d_{mle}):** the Levina–Bickel maximum likelihood estimate of local manifold dimension, capturing nonlinear structure that PCA may miss.
- **Trajectory alignment:** the mean cosine similarity between individual displacement vectors $(h_T - h_0)$ and the population mean displacement direction. High alignment indicates convergence toward a canonical reasoning pathway; low alignment indicates diffuse, exploratory dynamics.

- **Step-to-step coherence:** the mean cosine similarity between successive velocity vectors $v_t = h_{t+1} - h_t$ and v_{t+1}. Positive coherence indicates smooth flow; negative coherence indicates oscillatory "zig-zag" dynamics.

Additionally, we compute silhouette scores to quantify clustering into discrete modes, and the global-to-local dimension ratio (G/L) to measure manifold "tangling"—whether the representation is folded complexly through ambient space or lies comparatively flat.

1.3 From geometry to operators: the amortization hypothesis

Our geometric analysis is not merely descriptive. We propose an *amortization hypothesis*: if reasoning trajectories exhibit favorable geometry—low intrinsic dimension, high alignment, and clear clustering—then the endpoint of reasoning may be predictable without explicitly simulating every intermediate step. We define the endpoint operator

$$\mathcal{F}_c : h_0 \mapsto h_T \tag{2}$$

that maps initial hidden states to terminal hidden states under context c.

Unlike speculative decoding or draft-and-verify approaches—which remain fundamentally token-anchored and reduce constant factors in traversing a reasoning path—endpoint prediction targets a different axis entirely: whether the $O(T)$ sequential dependence can be bypassed at operator evaluation time when geometry permits.

To evaluate whether a predicted terminal state is *functionally* informative, we consider two decoding regimes. (i) **Frozen-unembedding decoding** applies the model's original, unmodified output head (unembedding) directly to $\hat{h}_T$ with no trained decoder, serving as a stringent diagnostic of whether the operator lands in the model's native coordinate frame. (ii) **Decoder-adapter (probe) decoding** trains a lightweight adapter to map predicted states to answer-token logits (Section 3.5.2). Unless stated otherwise, all reported "decoding accuracy" in this paper refers to the adapter/probe regime; frozen-unembedding decoding is treated as a diagnostic and is not used as the headline accuracy metric in our experimental setting.

1.4 Preview of findings: a phase diagram of thought

Our experiments reveal that reasoning geometry is not uniform across domains, and that scale induces domain-specific phase transitions rather than universal improvement. We identify three distinct geometric phases:

Crystallization (Law). Legal reasoning undergoes a dramatic structural transformation between 8B and 70B parameters. Global dimensionality collapses by 45% ($d_{95} : 501 \to 274$), trajectory alignment increases by 31% ($0.72 \to 0.94$), and the global-to-local ratio drops from $\sim 10\times$ to $\sim 1\times$, indicating complete manifold "untangling." The model does not merely reason *better* about law at scale—it reasons *differently*, compressing the space of possible reasoning pathways into a low-dimensional, highly reproducible structure. We term this phase *Crystalline*: rigid, ordered, and geometrically efficient.

Liquidity (Science, Math). In sharp contrast, scientific and mathematical reasoning exhibit scale invariance. Despite a $9\times$ parameter increase, global dimensionality remains unchanged (Science: $237 \to 235$; Math: $501 \to 501$), and manifold structure shows no significant reorganization. These domains maintain diffuse, exploratory geometry at both scales—a *Liquid* phase that resists compression. Scale provides more parameters but does not simplify the representational structure of scientific thought.

Lattice formation (Code). Programming tasks reveal a third phase characterized by discrete modularity. Code reasoning achieves the highest clustering quality